The causal structure and computational value of narratives

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Abstract

Many human behavioral and brain imaging studies have made use of narratively-structured stimuli (e.g. written, audio, or audio-visual stories) in an effort to emulate natural experience in the laboratory. However, evidence suggests that narratives are a special class of real-world experience, in large part defined by their causal connections across time, which much contemporary neuroscience research does not account for. Comprehension and memory for narratives are shaped by this causal network structure, as is activity in the brain systems which underlie their processing. Narrative structure in real-world events may increase the efficiency of planning and help us solve challenges in linking causes to outcomes in complex, high-dimensional environments.

Keywords

Narratives, causality, reasoning, value, credit assignment, plausibility

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Causal structure in narrative naturalistic stimuli

The use of "naturalistic" stimuli to study human brain function is motivated by the notion that they resemble real-world experience in terms of complexity, content, and statistics. This theoretically enables investigations of neural and mental phenomena that are unavailable when using more controlled and simplified stimuli. One thread of questions, mainly in the domain of vision science, has focused on how using artificial displays vs. natural images or videos affects neuronal response properties in both human and non-human animal subjects [1–3]. Related lines of research have focused on higher-level cognitive processes in the brain, such as semantic representations [4,5], memory [6–9], attention [10], and affect [11,12]; such studies require stimuli with cognitive complexity in addition to dynamic visual and auditory experience, and thus have tended to draw from media generated for entertainment, such as cinematic movies, television shows, podcasts, and literary fiction. These naturalistic stimuli are almost always "narratives", whether they are audio-visual, audio-only, or visual-only. A narrative may be defined, broadly, as a series of events [13], and recent cognitive neuroscience studies have been inspired by the classic psychological literature on event models [14,15]. However, while numerous studies have examined neural correlates of event perception and event segmentation during narrative processing, far fewer have addressed another key property of narratives which featured prominently in that same classic psychological literature: narratives are networks of *causally-connected* events, often forming chains of causal links, and these causal connections have a significant impact on comprehension and memory.

Recognizing the critical role of causal structure in narrative stimuli, and of causal reasoning in the processing of such stimuli, can enhance our understanding of prior results and guide the design of future experiments. It may also be possible for insights from studies of narratives to motivate questions in research on models of learning. For example, a key problem in reinforcement learning is the question of how to identify the specific prior actions or events which caused the current reward or punishment (**credit assignment**); this problem becomes especially challenging when the causal event occurred many events in the past (with many possible counterfactual alternatives along the way). In studies of narrative memory, causal connections between events are assessed by the reader based on their world knowledge of what types of actions may plausibly cause certain outcomes; e.g., if a king falls ill, a poisoned meal is a plausible cause while the arrival of a letter is not. Reinforcement learning models could be made more ecologically valid by incorporating prior knowledge about the plausibility of causal connections between classes of actions and outcomes, as such information is available for organisms when they make decisions in real-world scenarios. Beyond individual action-outcome pairs, the global causal structure of narratives is often familiar (“scripts”, Schank & Abelson 1995), which can provide information about the likelihood of causal connections across time; e.g., if a wife demands a divorce when her husband forgets to take out the trash, observers know that this minor incident is not the real cause, but that rather a number of prior events jointly contributed to her current decision. Models of learning could investigate how transferrable causal structure in these multi-event sequences is acquired across exposures where details may differ between instances. In return, a reinforcement learning perspective may be useful in helping elucidate the reasons for certain properties of narratives and why people seem to value them highly.

Accurate credit assignment is crucial to efficiently and effectively navigating a complex world, and narratives may constitute compact descriptions of the set of causal links which likely contributed to a given outcome (reward or punishment), extracted via inference and reasoning across past events and compiled into a form which is easy to encode into long-term memory, as well as communicable to others.

In this review, we survey behavioral and brain imaging studies which bear on the question of how causal structure in narratives shapes memory, highlighting investigations of neural systems which support episodic memory and mental simulation. In parallel, we draw links to work on causal attribution in studies of reinforcement learning, and suggest ways in which the two literatures may inform each other, with the goal of achieving a deeper understanding of the value of narratives.

Behavioral investigations of how causal structure in narratives impacts memory

Narrative comprehension was a highly active area of research beginning in the 1970s. These early studies used written narratives to investigate how readers build mental representations of events and what factors determine later memory. A key idea in this literature was that readers must go beyond the literal or surface features of text in order to generate meaningful representations of the events described in the story, and that doing so depends upon drawing on world knowledge; both surface form and meaning can be remembered, but they are distinct [16–19]. Researchers also examined how representation formation during narrative comprehension is guided by canonical schemas and meaningful relations among text elements (e.g., clauses or events), with more coherent representations yielding better memory performance [20,21].

A number of studies used networks to model mental representations of narratives. These studies found that causal relations between text elements or events (the nodes of the networks) are of critical importance in determining comprehension and memory. Narratives were typically divided into events, e.g., sentences or short passages (Figure 1A), and causal links between events were classified by experts (e.g., the researchers) according to a variety of logical criteria, such as temporal priority, operativity, necessity, and sufficiency [21–23]. In recent studies, causal links between events are more often identified by asking naïve participants for their “layman’s opinion” of whether a causal relationship is present (e.g.,[24]; Figure 1B). The causal structure of a narrative can then be visualized as an event-by-event matrix, either directed or undirected (1C), and as a network (1D). Some key findings from this literature were that creating a narrative from otherwise unrelated items dramatically improves recollection [25], the number of causal connections per event predicts judgments of importance and improved memory [21,22,26,27], and causal chains help organize event memory and determine what will be retained [28,29]. More recently, studies of how causal information affects memory for narratives have shown that recall is influenced by causal links more than temporal sequence [30], and that recall is enhanced for sentences which create causal breaks in a narrative [31]. In sum, causal connections are an essential determinant of how people understand and remember narratives [32].

Do causal connections in narratives imply causal reasoning in the course of narrative comprehension, and how does this compare to the types of reasoning that have been studied with simpler tasks? Theories of causal reasoning often define causal

relations between events according to how one event's existence affects the other [26]. For example, "if A occurs, B always occurs" we would say that "A causes B"; "if A occurs, B never occurs" we would say that "A prevents B"; both of these are causal relations between events. For simple physical, social, or biological scenarios, such relations can be studied in a highly controlled manner, e.g., asking participants to judge whether one ball caused another to miss a target [33], or what factors caused illness in a fictional world [34]. One possibility is that causal reasoning during narrative comprehension draws on the same mechanisms modeled in simpler tasks, combining them as needed depending on the content of the narrative events. However, the complexity of causal reasoning needed to comprehend popular-culture stories and movies (such as those used in naturalistic neuroimaging studies of cognition) seems to exceed the explanatory abilities of current models. For example, it is commonplace for a given event in a narrative to have multiple candidate causes (Figure 2), a problem which is challenging for formal models of reasoning. Additionally, it is unclear whether processing a narrative truly requires *acts* of causal reasoning at every event which is causally linked to a prior one; for example, if in Event 1 a wizard sends a letter to a king, and in Event 5 the king receives the letter, it seems unlikely that readers expend much effort pondering whether Event 5 depends on Event 1, or on what would have happened if Event 1 had never occurred; someone receiving a letter is simply a likely outcome of someone sending a letter. Yet, these two events would be strongly endorsed as being causally linked. Future work is needed to elucidate how causal reasoning during narrative comprehension is related to findings about causal reasoning processes in other domains and under simpler experimental conditions.

Overall, there is ample behavioral evidence that the pattern of causal connections between events in a narrative heavily influences its recollection. This privileged status of causal associations in memory retrieval supports efficient inferences, judgments, and planning [34–37]. Indeed, the preference for retrieving — and, by extension, encoding and maintaining — coherent causally-linked trajectories may be one way in which individuals can solve otherwise intractable problems of planning and credit assignment in complex, high-dimensional state spaces such as are encountered in the real world (Figure 2).

Neural correlates of processing causal structure in narratives

In terms of neural systems, there is limited work directly investigating the effects of causal structure on brain activity when people engage with or produce narratives. However, there are several reasons to suspect that narrative causal structure plays an important role in the activity of brain regions associated with episodic memory, e.g., default mode network (DMN) regions, medial temporal lobe (MTL) cortex, and the hippocampus.

The role of mental simulation in narrative comprehension

Causal reasoning might depend on constructing a mental simulation of predicted and counterfactual scenarios, and such simulation is linked to episodic memory systems. The mental model theory of reasoning [38] posits that comprehension of discourse entails constructing a mental simulation of possible scenarios given the currently described

event. Similarly, Bower & Morrow [39] describe their notion of a mental model or situation model in terms of simulation: "The bare text is somewhat like a play script that the reader uses like a theater director to construct in imagination a full stage production." While the authors were referring to readers comprehending text stimuli, the same idea could apply when the stimulus is a dynamic audiovisual movie: many aspects of the “stage production” are provided by the film, but many aspects are left to be generated by the audience. For example, in a scene where a woman is shown hurriedly packing a suitcase, the viewer might construct a mental model that she is running late for a train; if prior events have shown her ordering her deadbeat brother to move out, the viewer might construct a mental model that she is packing his things. Thus, even a rich audiovisual movie narrative is still a “bare” script in some sense, requiring the viewer to do substantial mental work in order to comprehend the depicted events.

Considerable evidence supports the idea that DMN cortical areas are involved in scene construction and imagery – that is, they may be the site of the "stage production" in the brain. fMRI studies have shown that imagining future events, e.g., visualizing a scene given a text cue such as “bus station”, elicits activity in medial prefrontal cortex, retrosplenial cortex, parahippocampal cortex, and the hippocampus [40]. The hippocampus is implicated in imagining scenes as well; hippocampal amnesics seem to have difficulty imagining vivid scenes [41,42], and hippocampal activity is elevated during both scene perception and construction [43]. In studies using narrative stimuli, it was shown that participants who listened to a verbal description of a cinematic audiovisual movie they had never seen before exhibited brain activity patterns in DMN regions that were similar, event by event, to those in participants who were actually watching the movie [44]; furthermore, DMN event patterns changed when previously seen movie events needed to be re-interpreted in light of new information [45]. In a study crossing different proposed features of situation models, posterior DMN regions (posterior medial cortex [PMC], angular gyrus [AG]) and parahippocampal cortex were found to carry information about the locations in which events occurred, anterior DMN (medial prefrontal cortex [mPFC]) coded for event schema, and perirhinal cortex and temporal pole coded for the people in the events, suggesting different contributions for these regions towards the construction of mental models during narrative comprehension [46]. These studies show how mental models of events could be represented in DMN activity patterns, perhaps populated in part by details retrieved from long-term memory via the MTL and hippocampus.

Neural substrates of association and simulation

Recent work points to a partial dissociation of representations for purely associative and causal structure. Specifically, a carefully designed experiment identified distinct neural correlates of predictions learned according to the Kalman Filter and Successor Representation rules, which reflect, respectively, predictive relationships and and long-range associative structure, and which were separately associated with activity in middle temporal gyrus and entorhinal cortex [48]. Critically, the middle temporal representation switched depending on the demands of the task, suggesting that individuals may flexibly rely on different horizons of temporal relationships when needed, consistent with the idea that narrative information supports next-step predictions based

on multiple scales of continuity. This dissociation is of potentially critical importance for understanding how narratives can support planning. Previous work identified planning-relevant representations in orbitofrontal cortex [49]; the rules that govern transfer between and arbitration among these representations have yet to be conclusively discerned. That each set of regions differentially engages with the DMN may indicate that the DMN supports the transfer of information between representations during offline processing, perhaps supporting the discovery of narrative structure [50,51].

Consistent with this view, several studies point to an overlapping network of regions being important in deliberative decision-making that relies on forward simulation. In particular, a medial prefrontal-hippocampal network has been shown to activate in proportion to the simulation demands of a decision problem [52,53]. In these studies, BOLD signal in this network was observed to increase when the anticipated outcome of a decision was more unpredictable – in other words, when the causal relationship between action and outcome was unclear. Both studies also further showed that such unclear situations yielded reinstatement of potential outcome representations in ventral temporal cortex, consistent with the use of mental simulation based on past experiences. This reinstatement activity also scaled with the difficulty of simulating next step outcomes, and, in a related study, was shown to correspond to the amount of activation in the hippocampus [54], consistent with the idea that the hippocampus orchestrates goal-directed retrieval of cortical patterns. A related study observed similar activation patterns in correspondence with the simulation demands of intertemporal choices [55], and that this activity (and the resulting decision) could be modified using episodic “tags” that cued the participant to simulate specific scenarios.

Brain systems which support episodic memory are influenced by narrative causal structure

A few recent studies have specifically investigated the role that causal connections might play in DMN and hippocampal responses during narrative processing. In one such study, participants were presented with temporally scrambled movies and reported moments of subjective comprehension ("Aha" moments); these occurred when participants identified causal connections between events. DMN activity was positively correlated with the subjective feelings of comprehension, and exhibited a more functionally integrated and efficient network state at these times [56]. In another study, researchers calculated the degree of causal connectedness ("causal centrality") of events in movies, calculated from the network of causal relations between all events, and assessed the impact on recollection and neural responses [24]. Unlike the classic behavioral studies mentioned above, here the causal networks were created without specific logical criteria, but rather by collecting judgments of causal linkage from naive participants. The results showed that responses in DMN regions (PMC and AG) scaled with causal centrality during spoken recollection of the movies, as well as during movie viewing (though in a less selective manner, not restricted to DMN areas). Furthermore, event-offset hippocampal responses were greater for more causally interconnected events, echoing the proposed idea that the hippocampus contributes to encoding memories at salient event boundaries [7,57].

Additionally, some studies which have investigated neural responses during narrative processing and memory have made observations relevant to causal structure,

even if the researchers' intentions did not focus explicitly on causal connections. In a study of hippocampal memory integration processes [58], brain activity was compared across pairs of temporally distant events that were "narratively coherent", i.e., they could be joined to create a two-event story; these event pairs were matched with other pairs which could not be logically joined into a story, and all were embedded in a longer unrelated narrative. Hippocampal activity patterns were more similar across the narratively coherent event pairs, and reinstated activity one day later was related to the detail of recollection. Interestingly, the "narratively coherent" event pairs in this study may be interpreted as being causally linked, and distinct in this respect from their matched counterparts. Other studies have explored the neural correlates of narrative "scripts", which can be thought of as higher-order categories of stories. For example, a pair of studies drawing from the same dataset showed that, when participants listened to or viewed stories taking place in restaurants and airports, the familiar series of causally linked events (e.g., looking at a menu, ordering food, eating, getting the check) could be parsed in DMN regions using a data-driven Hidden Markov Model algorithm, both during encoding [59] and recall [60]. In a study of how temporal information is tracked during a nonlinear narrative researchers showed that posterior medial DMN regions were sensitive to temporal jumps, and that inter-event neural similarity reflected chronological proximity moreso than the time of presentation within the narrative [61]. Though causal connections between events were not explicitly measured, the authors interpret their findings as relating to "our ability to embed the causal structure of events in a mental model".

Open questions remain as to whether causality-related computations during narrative comprehension and recall take place within the DMN or originate elsewhere, and whether retrospective judgments of causality by independent raters can provide an accurate basis for the timing of causal reasoning processes during narrative comprehension (see Box 3). Nonetheless, it is clear that narrative causal structure has a substantial impact on activity in the neural systems which support episodic memory, and as such it is important to understand this relationship and incorporate it into experimental designs when using narrative stimuli to study the brain.

A reinforcement learning account of the utility of narratives

Why are narratives treated preferentially in brain and behavior? One possible answer is that, by facilitating long-range associations between key actions across multiple events they may make it easier to perform important computations in support of value-based decision-making. In the framework of computational reinforcement learning, entry into states generates value signals (**Reward Prediction Errors**; RPE) proportional to the amount – and, critically, also the likelihood – of future reward that they entail. This signal is **discounted** in two senses: One, future rewards - those that can be expected to materialize only after one or more additional state transitions - are valued less than immediate rewards, usually according to a constant *discount factor* applied at each successive step. A second way in which distal rewards are discounted in value signals is via the estimated likelihood with which one can expect that reward to follow. A state that has many, relatively equally likely, potential next-steps, only one or a few of which lead to reward, will result in an attenuated value signal from a given future reward when

compared to a state with fewer potential successor states. Similarly, rewards seen to result only after a long chains of events contribute exponentially less to value signals relayed to the current state. This dispersion of value signals can be mitigated when information is acquired that allows the agent to increase their certainty of a particular next step or eventual outcome – in other words, when they have a higher expectation or certainty of a given successor state, as opposed to the others, the rewards that follow from that successor state will be more heavily credited to the current state. For example, if I walk into a restaurant that I know has one consistently satisfying dish, my expectation of future reward will be higher than when entering a restaurant where the quality varies between experiences.

This second form of discounting is where narrative structure can play a role. The framework implies that experiences that improve certainty about future rewards – for instance, by winnowing the tree of possible future states -- should themselves elicit value-like neural signals and value-like behaviors (attentional capture, preferential memory encoding), a prediction that has been borne out in several laboratory experiments over the past few decades. Perhaps the most striking of these phenomena was demonstrated in a study [62] wherein monkeys were offered a choice between two types of cues, each of which lead to the same chance of reward at the same delay, but one of which also lead to an intermediate cue that also signaled whether or not the trial would ultimately end in reward. Though the trials were perfectly matched in every other respect, they found that the monkeys showed a preference for the information-bearing option, and that the presentation of this option triggered a response in midbrain dopamine neurons reflecting the increased value – which they termed an “information prediction error.” Notably, the resulting prediction error at reward outcome was muted (or enhanced, if the information cue signaled less reward than actually appeared), reflecting an increase in certainty of the prediction of reward. This is one of several examples where humans and animals express a willingness to pay – in effort, material goods, or even foregone immediate reward – for mere information about the likelihood of upcoming rewards [63–65], which researchers have argued reflects the **value of information** provided by these passively observed cues [66]; one possible interpretation is that such information enables the animals to infer causality more quickly.

Narratively structured sequences provide such information at the outset, thus providing greater value of information than comparable stimuli. This fact may explain affective and behavioral responses attributable to information about rewards and their certainty. If one can use narrative schema to infer that the causal plausibility of one set of possible action sequences (for instance, tying my shoes is not likely to have any bearing on the outcome of my meal, whereas the choice of dish should) is more likely than another, value signals can spread more efficiently and correctly. This implies greater attention, fluency, and more precise and durable memory for such stimuli (see **Outstanding Questions** for more specific implications). Recent neural and behavioral findings add further support to this view. One functional imaging study examined the effect of plausibility on hippocampal activation during episodic counterfactual thinking, finding that increased plausibility yielded greater hippocampal activation, independent of effort and difficulty [67]. This finding is consistent with a preferential evaluation of narratively plausible stimuli. Another study presented individuals with multiple potential causal models to explain a given set of observations, finding that participants appear to show a

preference for the more compressed, or parsimonious, option [68], even when such models sacrifice some amount of overall accuracy. This result can be explained by the improved certainty of predictions of future events fitting with the given set of causal explanations.

This viewpoint can be useful not only in understanding the preference humans appear to have for narratives – especially compact, plausible ones – but also potentially in applying narrative schema to help improve our understanding of how reinforcement learning is made tractable for biological and artificial agents navigating complex, high dimensional worlds. Recent work [69–71] has shown that the firing of midbrain dopamine neurons, the neural substrate most closely associated with RPE in humans and animals, may (at least in some settings) be more precisely described as signaling an update to the contingencies associated with states that *precede* the current state, implying a predecessor representation akin to the successor described above. In other words, rather than encoding the *forward* prediction of discounted future reward, the signal conveys the *backward* prediction of states or features that causally lead to the present outcome – critically, even in the absence of explicit reward [71]. This finding is further supported by work in humans [36] showing that individuals preferentially leverage these backward predictions when a task's state space is *diverging*: Those tasks in which an agent moving forward in time can occupy more potential states than they have potential starting points. Interestingly, the reverse is also true – individuals are more likely to use forward planning when a task's state space is *converging*. Both of these findings are consistent with recent theoretical advances that show that it is advantageous for agents to construct compact representations of both backward and forward predictions to support efficient planning, especially in environments where rewards are unstable or state spaces are large but stationary [72].

Taken together, these findings support the idea that behavior and neural reward signals in humans and animals reinforce the acquisition and efficient representation of causal chains of events. We now turn to the question of why such sequences are treated as valuable, in and of themselves.

Credit assignment and the curse of dimensionality

A key problem to which these predictive representations are applied is **credit assignment** – identifying which of many potential actions or states is most likely to have yielded an encountered reward. Recent work has shown that efficient credit assignment benefits from selective counterfactual simulation, which is made possible by the sorts of backwards predictive representations described above [35]. However, the current state of the art algorithmic approach to generating such simulations is limited by the **curse of dimensionality** – when state spaces are large, it becomes difficult to prioritize a few out of very many possible preceding (or succeeding) sequences. In this setting, narrative schema could be thought of as a class of prior expectations about the likelihood of a given sequence of events, similar to the set of rich structure priors thought to support inference about the organization of natural categories [73]. This can enable two critical improvements: One, it allows an agent to ignore aspects of the environment that are not relevant to the likely space of tasks it will encounter – e.g. I do not need to expend representational resources encoding the likelihood that opening my closet door will result

in teleportation to a medieval forest. Second, it allows agents to select counterfactual simulations most relevant to the outcome at hand – when dissatisfied with my lunch, I most likely don't need to simulate what might have happened if I had chosen to wear different shoes. An agent endowed with rich priors about causal structure and plausible sequences – in other words, *narratives* – can use these to increase their certainty about the (forward and backward) contingency structure of their environment, which improves the efficient spread of value signals that support the selection of plans (Figure 3).

Concluding Remarks and Future Perspectives

Causal structure plays an important role in the way we process and remember narratives, both in terms of behavior and brain activity; consideration of causal variables can shed new light on prior studies and shape future experimentation. Many areas of cognitive research emphasize tasks and stimuli which have no causal structure across trials, and it is worth considering whether this omission leaves a critical gap in our understanding of how the mind and brain process the world. Furthermore, the utility that narratives hold for linking causes to outcomes in complex natural environments, and for making effective plans for the future, may help explain their seemingly privileged status in human episodic memory. These observations point to a number of avenues for future exploration of how causal structure in narratives, and in daily experience more generally, may influence human learning and health.

Impairments in reasoning are present in certain mental disorders, including schizophrenia and frontotemporal dementia. While schizophrenia is associated with disrupted causal perception and reasoning, dementia is more associated with problems in reasoning about psychological and social situations. Both disorders are marked by a decline in episodic memory. Interventions which make use of narratives and storytelling [74,75] may be effective in part due to the fact that recognizing causal connections is an integral aspect of narrative comprehension; emphasizing causal reasoning in the design of such interventions could increase their benefits.

Another way in which narratives may be valuable to mental health is in improving the adoption of adaptive mental actions, as in psychotherapeutic intervention for treating, e.g., rumination or substance use disorder. Hitchcock and Frank [76] showed that a major impediment to acquiring adaptive mental actions is that it is difficult to assign credit in mental action space, where many possible and imprecisely-defined states can follow a given goal directed action and precede the desirable outcomes. Narratively structured interventions, such as those that weave mental actions together with coherent descriptions of a self identity [77], should make credit assignment more efficient, thus perhaps improving treatment efficacy.

Humans and other animals depend on world knowledge, accumulated over their lifetimes, to identify cause-effect relationships between two specific events or moments in time. While the importance of such causal structure for processing and remembering narratives is established, it is still unclear how and when causal reasoning processes are engaged in relation to this network of cause-effect links. Furthermore, there may be distinct cognitive operations and neural substrates associated with processing different types of causal connections, e.g., physical, biological, psychological, social, structural; multiple of these are often present in the popular narratives used for modern research,

but their taxonomy is not well understood. Consideration of these questions may help bridge the gap between ideas about how humans identify and use causal information during complex multi-step experiences, such as narratives or daily life events, with studies and computational models of the neural mechanisms that enable organisms to link actions and outcomes.

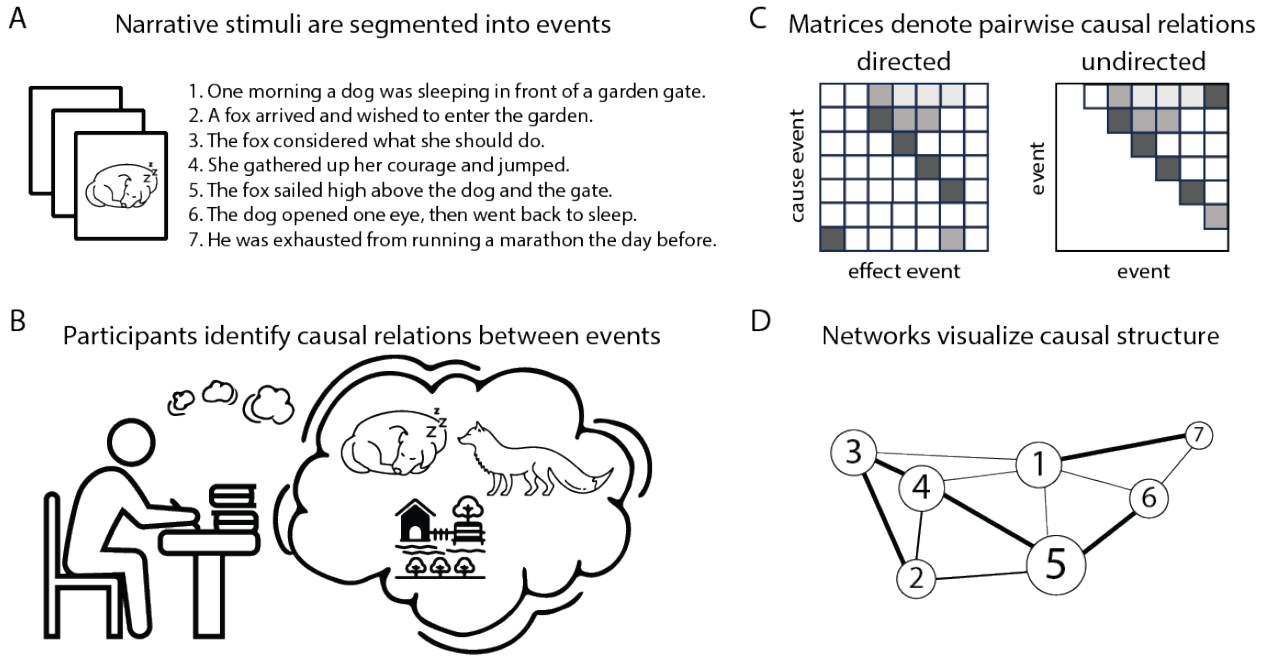

Figure 1. Creating and depicting narrative causal networks. **(A)** A narrative stimulus, such as a written story or a movie, is first divided into "events". In a written story, a sentence or a short passage typically corresponds to one event; in a movie, events may correspond to a short scene. In both cases, event boundaries are often defined by collecting judgments of event boundaries from a participant group [78]. **(B)** Judgments are collected from participants about the causal relationships between pairs of events [24,56]. **(C)** Pairwise relations between events can be plotted as directed or undirected matrices; directed matrices contain information about the direction of causality (which event caused which other event), while undirected matrices contain information about whether a link exists or not regardless of direction. The value in each cell indicates the strength of the connection, e.g., the proportion of the participants who identified that event pair as being causally linked. **(D)** Narrative causal event matrices can also be visualized as networks (undirected version shown here), wherein each event from the narrative is represented as a node. Node size indicates weighted degree centrality, i.e., the number and strength of connections, and edge thickness reflects the connection strength between events [24].

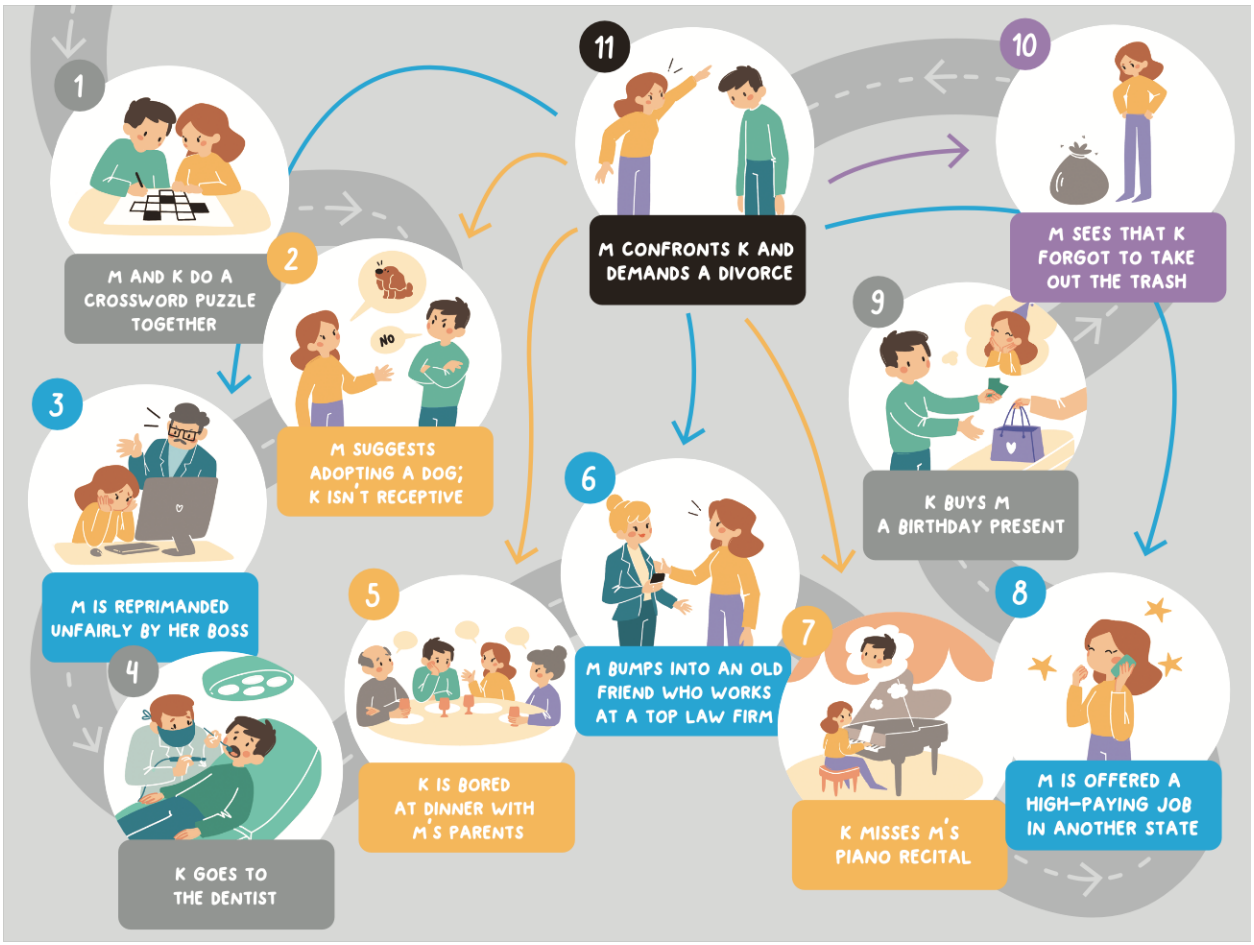

Figure 2. Assigning credit to narratively plausible prior events.

To which prior events should K assign credit for his current unwanted outcome (#11)? If he were to attribute the outcomes primarily to the immediately prior event (#10), he would come to the erroneous conclusion that M asked for a divorce because K forgot to take out the trash. Depending solely on temporal contingency to identify relevant states and actions could impede learning of the real causes. In contrast, world knowledge of what types of actions are plausible causes, and familiarity with narrative event sequences associated with romantic relationships, can help identify two sets of possible causes for event #11.

Narrative 1 ("**N**eglect"; orange): K neglected M's wishes and did not like her family or her hobbies; M's building resentment eventually caused her to snap.

Narrative 2 ("New **O**pportunity"; blue): M was unhappy at her job; a chance meeting with an old friend led to an attractive job offer in another state, and M realized that she wanted a fresh start without K.

Not all events in K's life contribute to solving the current credit assignment problem. The narrative sequences (blue and orange) are selected from a larger set which includes events (gray) that are not plausible causes of event #11. Using narrative structure to narrow and cluster these sets of possible causes can make efficient and effective credit assignment feasible, allowing the people involved to learn how to avoid situations leading to this outcome in the future. (Image credit: Anna Kuzmina)

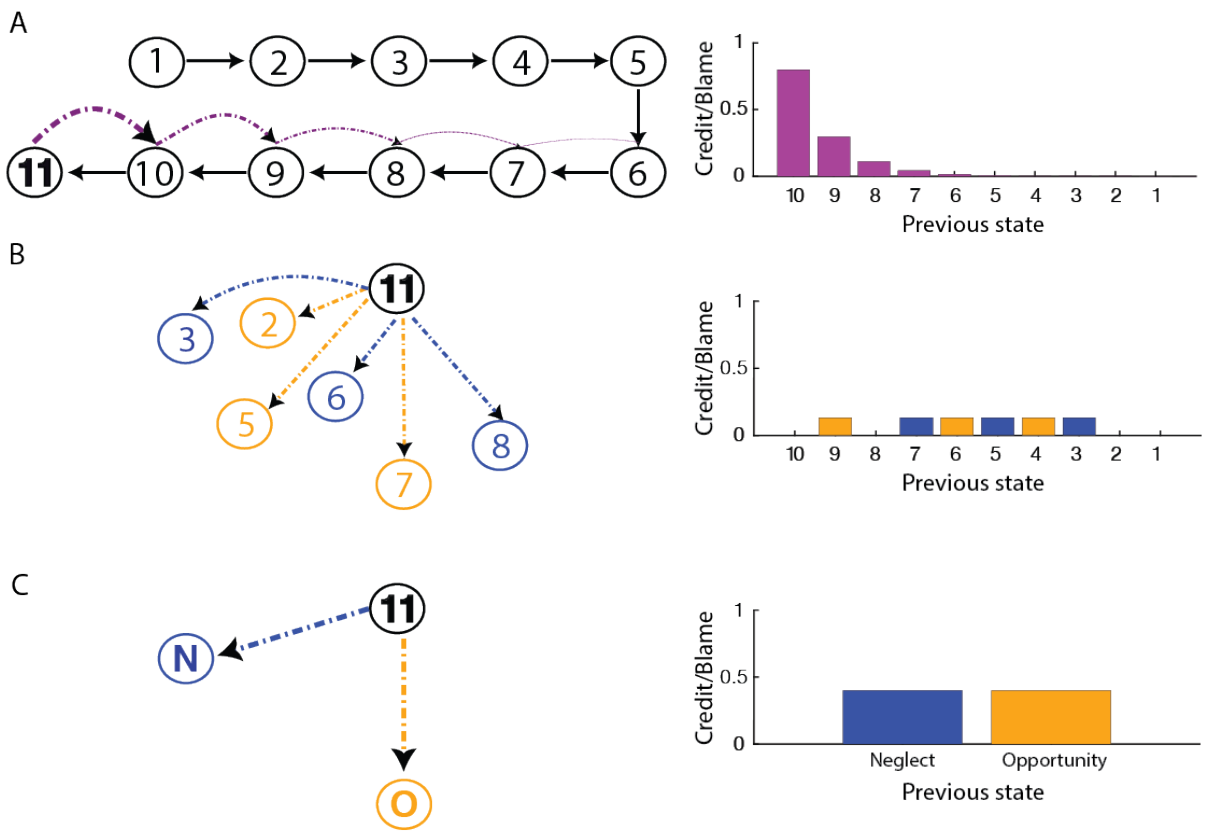

Figure 3. The effect of narrative structure on credit assignment.
We draw the example from Figure 2 as a state-space diagram to illustrate how narrative structure can improve the efficacy of credit assignment. **(A)** The chain of events that preceded the outcome of divorce (#11) is depicted in strict temporal sequence. Using this representation, an agent might assign credit ("blame") mostly the immediately preceding state (#10), with some decreasing credit assigned to preceding states as well. **(B)** Using causal plausibility to infer which of the states is responsible for the divorce highlights six states, which absent any further information can be assigned equal credit for the outcome. **(C)** The problem can be further simplified by realizing that each of the causally plausible events can be understood as one of two narratives that describe the evolution of the relationship. Now, credit can be assigned to each of these two narrative-states at a higher level of abstraction, guiding the agent to identify the potential outcomes of future entry into these states.

Box 1. How do we learn narrative plausibility?

Narrative plausibility depends on the world knowledge we have accumulated over a lifetime. When we are presented with a scene, the massive space of possible situations that the input (text, video, or the real-world scene before us) *could* be describing is instantly pruned by world knowledge of what is plausible. In other words, we have a plethora of latent assumptions that seem to guide our interpretations of events.

> *Parent describing a book they are reading: "After the man escaped from prison, he went into hiding for many years."*
>
> *Child: "What was he hiding behind?"*

To an adult, the parent's sentence immediately evokes a class of narratives (a schema) which involve a person concealing their identity, moving to a new city, and tense chase situations with government operatives. To the child, the sentence evokes something like a playground game of hide and seek, albeit a very long one, the oddness of which necessitates further questions.

The question of how world knowledge is acquired immediately arises, and indeed one wonders whether the burden of explanation has merely been shifted. It seems unnecessary to defend the existence of world knowledge and common sense; the focus of the current discourse is, rather, on how that world knowledge may be deployed at key moments to guide causal attributions and assign value accurately to past actions or events. Nonetheless, it is a critical open question how narrative plausibility is learned over the course of years, especially early in life.

Recent work shows that by age 3, childrens' behavior is sensitive to narrative structure in fictional stories [79]. Higher-order associations undergo a sensitive period, during which experiencing unpredictable contingencies in the environment may permanently alter mechanisms for interpreting the causal relationships between events [80], with impacts on learning, decision-making, and exploration [81–84]. New, publicly available narrative movie fMRI and EEG datasets with child participants [85,86] also open the door to exploration of developmental questions [87,88].

Box 2. Is there a link between temporal receptive windows and narrative causal coherence?

An influential line of research made use of naturalistic stimuli to show that information processing timescales ("temporal receptive windows") vary regularly along the cortical hierarchy [89–91]. In this hierarchy, the shortest timescales are observed in low-level sensory areas, and the longest timescales are observed in higher-order regions more distant from the sensory periphery – overlapping substantially with the DMN. These timescales were measured using temporal scrambling of naturalistic stimuli (both movies and stories). In DMN cortical areas, responses were affected even when the input was scrambled minimally: randomly re-ordered at the event level (in chunks as long as 30-60 seconds). The key result from these studies is that timescales increase systematically across the brain, a principle which cuts across sensory modalities; thus, it was never

specified what kind of information was being disrupted by the scrambling. In retrospect, a possible explanation is that event-level scrambling specifically affected the integrity of causal structure in the narrative stimuli. In a narrative with randomly re-ordered events, it may be difficult or impossible to identify causal links between events, leading to impaired comprehension and decreased synchrony across individuals (synchrony being the main measure of interest in these studies). Future work could examine whether the breakage of causal connections in narratives is indeed the key factor that determines whether DMN activity patterns de-synchronize across individuals. Such a finding might suggest that the information accumulation function of long-timescale brain areas subserves the identification of causal connections across temporally extended events.

Box 3. Interpreting extant findings about how narrative causal structure impacts brain activity

It is well known that DMN activity is correlated with retrieval success and reactivation of episodic memories. In studies using narrative stimuli, is it possible to disentangle whether DMN responses reflect 1) their involvement in causal reasoning processes during encoding and retrieval, as opposed to 2) enhanced memory encoding and retrieval as a consequence of causality-related computations occurring elsewhere in the brain? One argument against the latter view is that, while DMN activity is positively related to retrieval success, there are numerous studies showing that it is lower during encoding for items which are subsequently remembered [92,93]. However, studies showed that DMN activity was *greater* during encoding when causal recognition [56] or causal connections [24] were higher, and corresponded to *higher* likelihood of subsequent recall for these events [24]. These observations do not fit neatly with an account of DMN responses to causal processing being merely reflective of enhanced encoding. Meanwhile, studies using non-narrative stimuli have yielded mixed answers to the question of which brain regions support causal reasoning processes, variously implicating networks associated with attention, memory, and executive control (for review, see [94]). To the extent that causal reasoning depends on amodal mental simulations of current and possible events [38] – which, intuitively, may be more true for narrative and daily-life use cases – a key role for the DMN in acts of causal reasoning seems like a plausible proposition.

A related question is whether causal connections in a narrative, when identified retrospectively by expert or naive judgment (Figure 1), provide information about the *moments in time* at which readers or viewers are engaging in causal reasoning processes as they experience the narrative. For instance, many narratives have a prominent chain structure, in which almost every event is judged to have a causal relationship with temporally adjacent events (both immediately prior and immediately following). If every such causal link in a narrative requires causal reasoning, that would imply that these processes are engaged almost continuously during narrative comprehension. Another possible disconnect between retrospective ratings and ongoing causal reasoning is that judgments about events early in a story may be altered in light of information arriving in later parts of the story; this effect could be much greater for certain types of narratives, such as those with surprise endings [45,95] or with intentionally disrupted temporal order [56,61]. Yet another disconnect could arise from individual variability in extracting causal

properties of a narrative; for example, in a murder mystery, some viewers might guess the identity of the killer much earlier than others.

Glossary

- **State space**: An abstraction used in reinforcement learning to distinguish configurations or situations possible in a given environment. States can be defined at multiple levels of abstraction: An ice cream store may be described as a single state, or it may be multiple states that vary with what's in stock. The level of abstraction is chosen to match the goals of the learning agent – more detailed state-spaces can lead to more nuanced policies, at the cost of greater representational and computational complexity.
- **Reward Prediction Errors:** Upon entry into a state, the difference between the value expected and that observed. Critically, in reinforcement learning models, the value signal is the (discounted) sum of all future rewards that could be expected to follow, until the termination of the task episode.
- **Value of information:** The increase in future expected value that results from obtaining a given piece of information. This quantity can result from an increase in the likelihood of a given future reward, or an increase in certainty about the delivery of rewards.
- **Credit assignment:** The problem of selecting which among many preceding actions or states should be assigned value crediting it for being predictive of a given experienced reward.
- **Curse of dimensionality:** Broadly, many different issues that arise when working with large, high-dimensional data that do not appear in lower dimensions. For the purposes of this article, we use this term to refer to the difficulty of credit assignment in large, naturalistic environments with many dynamic features and action sequences.
- **Discounting:** The degree to which future rewards are valued less than present rewards. This factor can be justified by uncertainty about the receipt of future rewards, or by their lesser utility with respect to present goals.
- **Successor representation:** A compact representation of the likelihood that trajectories beginning from each possible state in a given environment will eventually occupy each of the other possible states; these can be computed either at the full horizon of the task (e.g. extending to the terminal state), or at multiple intermediate timescales, giving *n*-step successor representations.

Citation diversity statement

Recent work in several fields of science has identified a bias in citation practices such that papers from women and other minority scholars are under-cited relative to the number of such papers in the field [96–104]. Here we sought to proactively consider choosing references that reflect the diversity of the field in thought, form of contribution, gender, race, ethnicity, and other factors. First, we obtained the predicted gender of the first and last author of each reference by using databases that store the probability of a first name being carried by a woman [100,105]. By this measure (and excluding self-citations to the first and last authors of our current paper), our references contain 18.22% woman(first)/woman(last), 18.35% man/woman, 16.26% woman/man, and 47.18% man/man. This method is limited in that a) names, pronouns, and social media profiles used to construct the databases may not, in every case, be indicative of gender identity and b) it cannot account for intersex, non-binary, or transgender people. Second, we obtained predicted racial/ethnic category of the first and last author of each reference by databases that store the probability of a first and last name being carried by an author of color [106,107]. By this measure (and excluding self-citations), our references contain 7.18% author of color (first)/author of color(last), 16.08% white author/author of color, 18.32% author of color/white author, and 58.42% white author/white author. This method is limited in that a) names and Florida Voter Data to make the predictions may not be indicative of racial/ethnic identity, and b) it cannot account for Indigenous and mixed-race authors, or those who may face differential biases due to the ambiguous racialization or ethnicization of their names. We look forward to future work that could help us to better understand how to support equitable practices in science.

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